Question 2: Spotting low-quality transcribers

Product case study · Swapnil Ghatak · October 2026. Dataset from a crowdsourced transcription platform (not included).

Summary

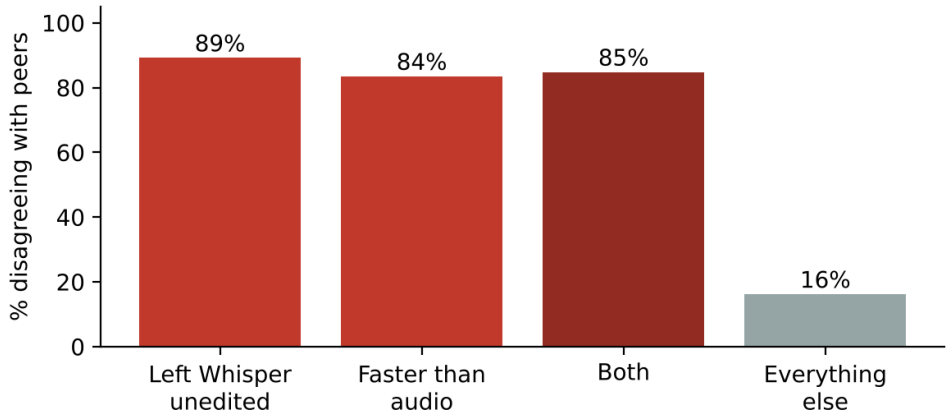
I looked at 51,783 transcription tasks. The clearest warning sign is someone leaving Whisper's text exactly as it is while spending less time on the task than the audio itself lasts. In other words, they couldn't have listened to the whole clip. Some clips were transcribed by several people, and on those, 85% of these submissions disagreed with what everyone else wrote, against 16% for all other submissions. The brief asks to be 100% sure before blocking anyone, and no data signal can give that on its own. So I'd use the signals to decide who gets checked more closely, and only block someone after they fail hidden test clips with known answers and a person has reviewed their work.

Three things I noticed about the data first

- **Only 10 user IDs appear more than once.** Between them they have 2,918 rows (86 to 609 tasks each). The other 48,865 IDs appear exactly once, so I can only look at individual behaviour for these 10. I used all the other rows as a baseline for what normal looks like.
- **segment_character_per_second is based on audio length, not time taken.** The brief describes it as characters divided by time taken, but when I recalculated it, characters divided by audio duration matched 99.4% of rows, and characters divided by time taken matched only 0.1%. So it tells you how much text there is per second of audio, not how fast someone typed.
- **223 clips were transcribed by more than one person** (5,532 submissions). That gave me something close to a right answer: if one person's text is very different from what everyone else wrote for the same clip, it's probably wrong. I used this to test whether each warning sign actually points to bad work. I measured agreement as text similarity after removing spaces and punctuation, and counted anything under 0.8 as disagreeing.

Part I: Warning signs in the data

Checking the signals on the shared clips: submissions that show the first two signals disagree with other transcribers about five times as often as everything else.



Signal 1: Rubber-stamping without listening

What it tells us	The person kept Whisper's text unchanged and submitted before the audio could have finished playing, so they can't have listened to all of it. Any mistake Whisper made goes straight into the dataset, and Whisper makes plenty: 84% of all tasks needed edits.
How to measure it	Per task: is_edited = No AND whisper_text not empty AND time_taken ÷ duration < 1. Per user: share of tasks where this is true, over at least 30 tasks.
Red-flag threshold	Baseline is 7.1% of tasks. Flag a user at 10% or more. Strong concern at 20% or more (User A is at 24%).
False alarms	Playback speed-up (1.5x/2x) if the platform offers it; very clean audio where Whisper is genuinely right; very short clips. That is why it is measured as a rate over many tasks, not per task.

Signal 2: Time spent does not depend on audio length

What it tells us	Someone doing the work properly needs more time for a 15-second clip than a 2-second one. If they spend about the same time on every clip whatever its length, they probably aren't listening.
How to measure it	Per user: rank correlation (Spearman) between duration and time_taken (excluding idle times over 10 minutes); and share of clips longer than 6 s finished faster than the audio.
Red-flag threshold	Baseline correlation is 0.55 and 20% of long clips are finished faster than audio. Flag correlation below 0.2 AND over 50% of long clips faster than audio. All 10 tracked users show correlation near 0 and 75–94% faster-than-audio.
False alarms	A logging problem. All 10 tracked users have whole-number times spread evenly between 4 and 12 seconds, while everyone else has fractional times. That looks like it could be a tracking issue rather than real behaviour, so this signal should never be enough to block anyone by itself.

Signal 3: Keeping Whisper's hallucinations

What it tells us	Sometimes Whisper gets stuck and repeats one word for the whole clip. Anyone who listened would notice straight away, so submitting it unchanged is about as clear a sign as you can get that they didn't listen.
How to measure it	Per task: one word makes up more than 50% of an 8+ word whisper_text, and the user_text is still the same loop. 33 such submissions in the data.
Red-flag threshold	Even 1 occurrence on a known-hallucination clip is a strong flag; 2 or more is grounds for review.
False alarms	The speaker really did repeat a word (rare, and checkable on the audio).

Supporting signal: Cosmetic edits and inconsistent tags

What it tells us	Changing only a comma or a space sets is_edited to Yes without fixing anything, which can hide someone who isn't really checking. Separately, writing the blank tag many different ways shows the guidelines aren't being followed.
How to measure it	Share of edits where the text is identical after removing punctuation and spaces; number of different blank-tag variants a user writes.
Red-flag threshold	Cosmetic edits above 5% (baseline 1.9%). More than 3 tag variants (one user wrote 21).
False alarms	Fixing punctuation is a normal part of the job, and inconsistent tags are more about training than cheating. I'd use this for coaching, not blocking.

What I'd conclude about the 10 users

User	Tasks	Unedited when Whisper had text	Rubber-stamp unheard	Long clips faster than audio	Verdict
Baseline	48,865	31%	7.1%	20%	—
User A	165	78%	23.6%	78%	Strongest case: all signals agree, plus a kept hallucination. Gold-check first.
User B	86	32%	14.0%	94%	Second tier: rubber-stamps above baseline, extreme speed.
User C	609	25%	3.8%	80%	Second tier: speed anomaly, kept a hallucination.
Other 7	149–509	0–16%	0–2.3%	75–81%	Speed anomaly only; they edit almost everything. Could be diligence or fake edits. Gold-check, do not block.

Part II: How I'd detect and block automatically

Blocking someone cuts off their income, so we need to be sure. None of the data signals can make us sure by themselves: the timing pattern might be a logging bug, and on very clean audio an honest person might rightly leave Whisper's text alone. What can make us sure is **gold clips**: clips where we already know the correct transcript, mixed into a person's normal work without them knowing. So the data signals decide who gets checked more, and the gold clips decide who gets blocked.

Stage	Trigger	Action
0. Always on	Every transcriber	About 1 in 50 tasks is a gold clip. Score = word accuracy against the verified transcript.
1. Watch	After 30+ tasks: rubber-stamp-unheard 10% or more, OR time-vs-length correlation below 0.2 with 50%+ long clips faster than audio, OR any kept hallucination	Raise gold rate to 1 in 10. No impact on the user.
2. Warn and hold	Gold word accuracy below the platform target (e.g. 95%) on 3+ gold clips	Hold payment for flagged tasks. In-app message showing their gold mistakes and the guideline. Short re-training quiz.
3. Block	Gold accuracy still below target on 5+ further gold clips after the warning, AND at least one Stage-1 signal, AND a human reviewer agrees after listening to 5 tasks	Block account. Unpaid flagged tasks re-queued. User can appeal once.
Instant review	Unedited hallucination loop on a gold clip	Skip to human review; block if confirmed.

As a single rule

If a transcriber (X) still scores below target on gold clips after being warned, and (Y) shows at least one of the warning signs over 30 or more tasks, and (Z) a reviewer listening to their work agrees, then block them. X on its own means retraining, Y on its own means more gold clips, and neither on its own leads to a block.

What I'd ask the product and engineering team for

- **Better time tracking.** Log when the audio is played, paused, skipped or sped up, and how long someone is actually editing, not just the time from opening to submitting. Right now the time field can't tell a quick honest worker from someone who didn't listen, or from a broken timer.
- **Make people play the audio before submitting** on clips longer than 3 seconds (say, at least 80% of it). That stops the first problem from happening instead of catching it afterwards.
- **Tag buttons** for [blank] and [noise], instead of letting people type them.
- **Show people their own score on gold clips.** A lot of poor work can be fixed with feedback, and it keeps good contributors motivated.
- **Set the thresholds using real data** from the first month of gold clips before switching blocking on, and keep track of how many blocks are overturned on appeal.

All numbers in this document come from q2_transcriber_quality.py (Python, pandas).